Lab 05 – Worksheet

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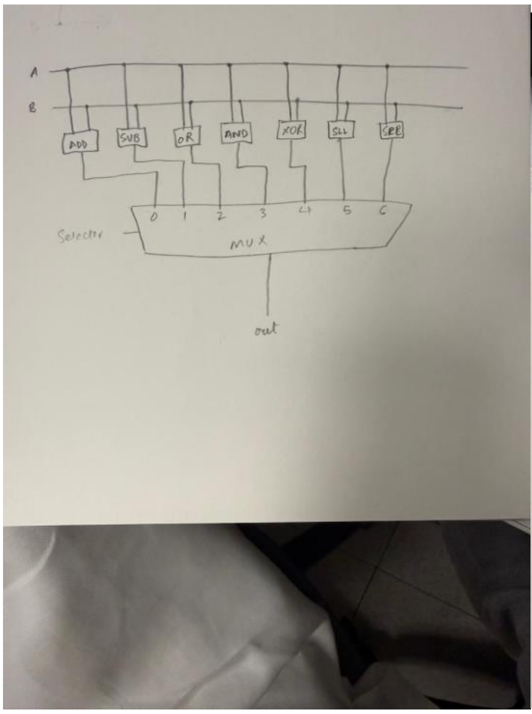
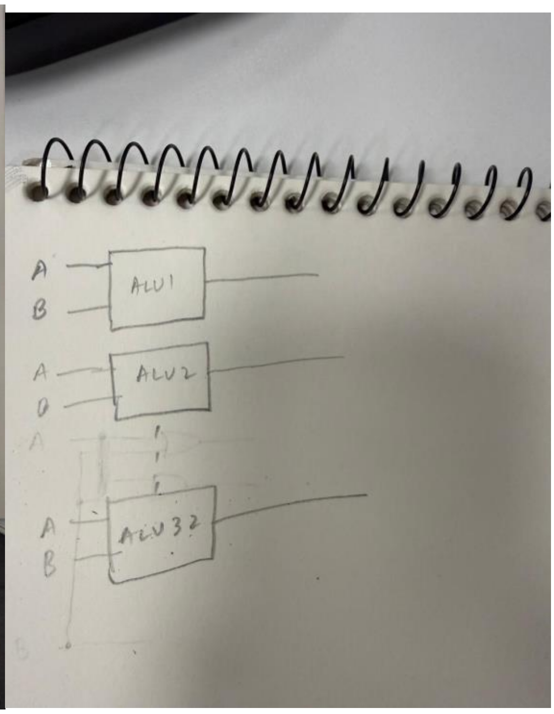
Task 1

Provide appropriately commented codes.





Task 2

Provide appropriately commented codes

ALU.v:

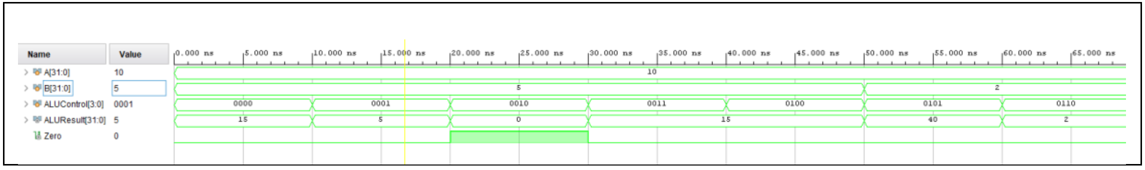
```
module ALU(input [31:0] A, input [31:0] B, input [3:0] ALUControl, output reg
[31:0] ALUResult, output Zero);
always @(*)
   begin
      case(ALUControl)
         4'b0000: ALUResult = A + B;
         4'b0001: ALUResult = A - B;
         4'b0010: ALUResult = A & B;
         4'b0011: ALUResult = A | B;
         4'b0100: ALUResult = A ^ B;
         4'b0101: ALUResult = A << B[4:0];
         4'b0110: ALUResult = A >> B[4:0];
         default: ALUResult = 32'b0;
      endcase
   end
assign Zero = (ALUResult == 0);
endmodule
```

ALU_tb.v:

```
module ALU_tb;

reg [31:0] A;
reg [31:0] B;
reg [3:0] ALUControl;
wire [31:0] ALUResult;
wire Zero;

ALU alusahab(A, B, ALUControl, ALUResult, Zero);

initial
   begin
   // adding
   A = 10; B = 5; ALUControl = 4'b0000;
   #10;
   // subtracting
   A = 10; B = 5; ALUControl = 4'b0001;
   #10;
   // anding
   A = 10; B = 5; ALUControl = 4'b0010;
   #10;
   // oring
   A = 10; B = 5; ALUControl = 4'b0011;
   #10;
   // xoring
   A = 10; B = 5; ALUControl = 4'b0100;
```

```
    #10;
    // slling
    A = 10; B = 2; ALUControl = 4'b0101;
    #10;
    // srling
    A = 10; B = 2; ALUControl = 4'b0110;
    #10;
    end
endmodule
```

Add screenshot of your results



Comments

As we can see our A is 10 that is 1010 and B is 5 that is 0101 and then we perform all operations on it show in the waveform

Task 3

Provide appropriately commented codes

ALU.v:

```
module ALU(input [31:0] A, input [31:0] B, input [3:0] ALUControl, output reg
[31:0] ALUResult, output Zero);
always @(*)
    begin
        case(ALUControl)
            4'b0000: ALUResult = A + B;
            4'b0001: ALUResult = A - B;
            4'b0010: ALUResult = A & B;
            4'b0011: ALUResult = A | B;
            4'b0100: ALUResult = A ^ B;
            4'b0101: ALUResult = A << B[4:0];
            4'b0110: ALUResult = A >> B[4:0];
            default: ALUResult = 32'b0;
        endcase
    end
assign Zero = (ALUResult == 0);
endmodule
```

leds.v:

```
module leds(
   input clk, rst,
   input [15:0] btns,
   input [31:0] writeData,
   input writeEnable,
   input readEnable,
   input [29:0] memAddress,
   input [15:0] switches,
   output reg [31:0] readData
);
   always @(posedge clk) begin
      if (rst) readData <= 0;
      else     readData <= {16'b0, switches};
   end
endmodule
```

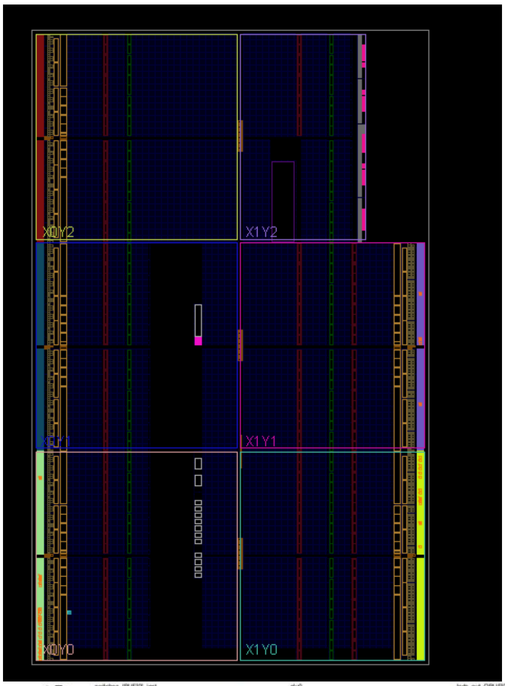
switches.v:

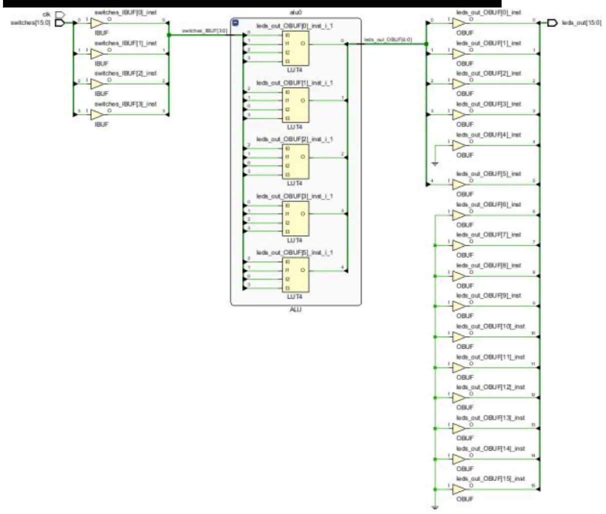
```
module switches(
   input clk,
   input rst,
   input [31:0] writeData,
   input writeEnable,
   input readEnable,
   input [29:0] memAddress,
   output reg [31:0] readData = 0, // not to be read
   output reg [15:0] leds
);
   always @(posedge clk) begin
      if (rst) begin
         leds <= 0;
         readData <= 0;
      end else begin
         readData <= {16'b0, leds};
      end
   end
endmodule
```

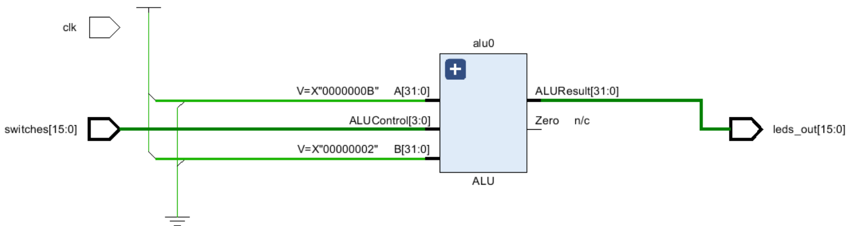
top_ALU.v:

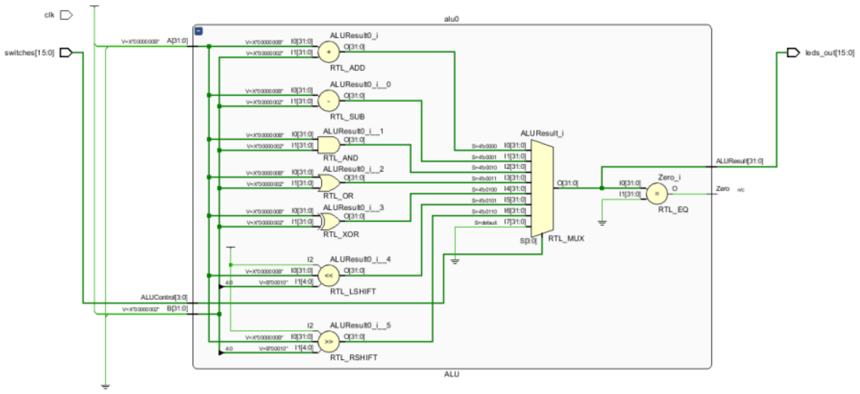
```
module top_alu(input clk, input [15:0] switches, output [15:0] leds_out);
   wire [31:0] A = 32'b00000000000000000000000000001011;
   wire [31:0] B = 32'b00000000000000000000000000000010;
   wire [3:0] ALUControl = switches[3:0];
   wire [31:0] ALUResult;
   wire Zero;
   ALU alu0(.A(A), .B(B), .ALUControl(ALUControl), .ALUResult(ALUResult),
.Zero(Zero));
   assign leds_out = ALUResult[15:0];
endmodule
```

Add screenshots of your results









Demo shown in lab on fpga board to miss maham.